1. Smart Actuated Valve with Predictive Maintenance

Research Gap:

- Most low-cost actuated valves only provide open/close control.
- Limited integration of condition monitoring and predictive maintenance features in affordable valve systems.
- Lack of early fault detection for valve sticking, wear, and actuator degradation.

2. Self-Balancing Two-Wheel Robot

Research Gap:

- Most low-cost self-balancing robots are tested only on smooth surfaces.
- Stability and control performance on uneven terrain remain insufficient.
- Limited research on improving balancing algorithms using inexpensive sensors.

3. Self-Balancing Cube

Research Gap:

- Most existing self-balancing cubes are expensive research platforms.
- Limited studies on low-cost implementations using readily available components.
- Challenges remain in improving balance duration and disturbance rejection.

4. Intelligent Solar Panel Tracking System

Research Gap:

- Existing tracking systems often consume significant power.
- Limited research on maximizing net energy gain while minimizing tracking energy consumption.
- Need for affordable and reliable tracking systems for small-scale solar installations.

5. Robotic Pipe Inspection Vehicle

Research Gap:

- Commercial inspection robots are expensive and inaccessible to many organizations.
- Limited development of low-cost inspection robots for small-diameter pipelines.
- Navigation and defect detection in complex pipe networks remain challenging.

6. Automated Irrigation Valve System

Research Gap:

- Many irrigation systems rely on fixed schedules rather than real-time field conditions.
- Limited integration of soil moisture data with automated valve control.
- Need for cost-effective water conservation solutions.

7. Active Suspension Test Rig

Research Gap:

- Active suspension systems remain costly for widespread adoption.
- Limited studies comparing low-cost active and passive suspension systems.
- Need for affordable experimental platforms for suspension optimization.

8. Smart Pneumatic Gripper

Research Gap:

- Conventional pneumatic grippers often apply fixed gripping forces.
- Limited development of adaptive force control for handling fragile objects.
- Need for low-cost intelligent gripping solutions.

9. Autonomous Waste Collection Robot

Research Gap:

- Existing systems focus primarily on navigation rather than collection efficiency.
- Limited studies on optimizing waste collection mechanisms.
- Need for affordable autonomous waste management solutions.

10. Stair-Climbing Inspection Robot

Research Gap:

- Most inspection robots struggle with stair navigation.
- Existing stair-climbing mechanisms are often complex and expensive.
- Need for energy-efficient climbing systems.

11. Automatic Pothole Detection and Mapping Robot

Research Gap:

- Current road monitoring methods are labor-intensive and costly.
- Limited availability of low-cost autonomous road inspection systems.
- Need for improved pothole detection accuracy.

12. Robotic Weed Removal System

Research Gap:

- Heavy dependence on chemical herbicides remains a concern.
- Limited affordable robotic solutions for small-scale farmers.
- Need for improved weed recognition and removal efficiency.

13. Smart Conveyor Sorting System

Research Gap:

- Small industries often lack access to automated sorting technologies.
- Existing systems may be costly and difficult to customize.
- Need for low-cost intelligent sorting solutions.

14. Automated Material Handling Robot

Research Gap:

- Material transportation remains labor-intensive in many facilities.
- Existing autonomous systems are often expensive.
- Need for cost-effective navigation and load-handling solutions.

15. Robotic Fire Detection Vehicle

Research Gap:

- Fire inspection in hazardous environments remains risky for personnel.
- Limited development of affordable autonomous fire-monitoring robots.
- Need for improved fire source detection and localization.

16. Smart Wheelchair with Obstacle Avoidance

Research Gap:

- Smart wheelchairs are often expensive and inaccessible.
- Existing systems may have limited obstacle detection capabilities.
- Need for affordable assistive mobility technologies.

17. Adaptive Robotic Gripper

Research Gap:

- Conventional grippers are designed for specific object shapes.
- Limited capability to adapt to varying object geometries.
- Need for versatile and low-cost adaptive gripping mechanisms.

18. Automatic Window Cleaning Robot

Research Gap:

- Existing systems face challenges in adhesion and movement on vertical surfaces.
- High costs limit adoption.
- Need for safer and more energy-efficient cleaning mechanisms.

19. Autonomous Warehouse Robot

Research Gap:

- Warehouse automation remains expensive for small businesses.
- Limited development of low-cost navigation systems.
- Need for efficient path-planning and obstacle avoidance strategies.

20. Active Vehicle Anti-Rollover System

Research Gap:

- Existing anti-roll systems are mainly used in high-end vehicles.
- Limited affordable solutions for commercial and passenger vehicles.
- Need for improved rollover prediction and prevention techniques.

21. Smart Shock Absorber System

Research Gap:

- Traditional shock absorbers cannot adapt to changing road conditions.
- Semi-active systems remain relatively expensive.
- Need for affordable adaptive damping technologies.

22. Active Vibration Control Platforms

Research Gap:

- Passive vibration control may not adequately suppress dynamic disturbances.
- Existing active systems can be costly and complex.
- Need for low-cost active vibration suppression solutions.

23. Automated Water Distribution Valve Network

Research Gap:

- Water distribution systems often suffer from leakage and inefficiency.
- Limited implementation of intelligent flow control in low-cost systems.
- Need for automated water management solutions.

24. Smart Tank Level Control System

Research Gap:

- Manual monitoring of storage tanks remains common.
- Existing systems may not optimize pump operation.
- Need for reliable and affordable level monitoring and control.

25. Automatic LPG Leak Detection and Isolation System

Research Gap:

- Gas leaks continue to cause accidents and property damage.
- Many systems only provide alarms without automatic isolation.

- Need for rapid-response leak detection and shutoff mechanisms.

26. Low-Cost Prosthetic Hand

Research Gap:

- Advanced prosthetic hands remain expensive.
- Limited access to affordable prosthetics in developing countries.
- Need for functional and locally manufacturable designs.

27. Active Vibration Control Platform

Research Gap:

- Passive vibration reduction techniques have performance limitations.
- Existing active systems are expensive and difficult to implement.
- Need for affordable and energy-efficient vibration control technologies.